iii. Protons are transported through the membrane into nearby subsidiary cells with the exchange of K^+ (Potassium ions) from subsidiary cells to guard cells. This process involves an electrical gradient and is called **ion exchange**.

iv. This ion exchange is an active process and consumes ATP for energy.

v. Increased K^+ ions in the guard cell are balanced by Cl^- ions. Increase in solute concentration decreases the water potential in the guard cell.

vi. Guard cell becomes hypertonic and favours the entry of water from surrounding cells.

vii. Increased turgor pressure due to the entry of water opens the stomatal pore (Figure 11.16).

In Dark

i. In dark, photosynthesis stops and respiration continues with accumulation of CO_2 in the sub-stomatal cavity.

ii. Accumulation of CO_2 in cell lowers the pH level.

iii. Low pH and a shortage of water in the guard cell activate the stress hormone **Abscisic acid** (ABA).

iv. ABA stops further entry of K^+ ions and also induce K^+ ions to leak out to subsidiary cells from guard cell.

v. Loss of water from guard cell reduces turgor pressure and causes closure of stomata (Figure 11.17).

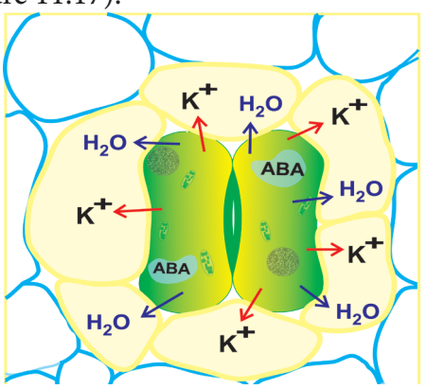


Figure 11.17: Theory of K^+ transport

Closing of stomata

11.6.4 Factors Affecting Rate of Transpiration

The factors affecting the rate of transpiration can be categorized into two groups. They are 1. External or Environmental factors and 2. Internal or plant factors.

1. External or Environmental factors

i. Atmospheric humidity: The rate of transpiration is greatly reduced when the atmosphere is very humid. As the air becomes dry, the rate of transpiration is also increased proportionately.

ii. Temperature: With the increase in atmospheric temperature, the rate of transpiration also increases. However, at very high-temperatures stomata closes because of flaccidity and transpiration stop.

iii. Light: Light intensity increases the temperature. As in temperature, transpiration is increased in high light intensity and is decreased in low light intensity. Light also increases the permeability of the cell membrane, making it easy for water molecules to move out of the cell.

iv. Wind velocity: In still air, the surface above the stomata get saturated with water vapours and there is no need for more water vapour to come out. If the wind is breezy, water vapour gets carried away near leaf surface and DPD is created to draw more vapour from the leaf cells enhancing transpiration. However, high wind velocity creates an extreme increase in water loss and leads to a reduced rate of transpiration and stomata remain closed.

Activity

What will happen if an indoor plant is placed under fan and AC?

v. Atmospheric pressure: In low atmospheric pressure, the rate of transpiration increases. Hills favour high transpiration rate due to low atmospheric pressure. However, it is neutralized by low temperature prevailing in the hills.

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